Swiss Cheese Model

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Introduction

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Product Owner, Connected Care

Lilly Diabetes and Obesity

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- BS in Biomedical Engineering
- MS in Biomedical Engineering
- MS in Systems Engineering (focus in Human Factors)

- 10+ years in medical devices, 4+ years in human factors

Topics

- Swiss Cheese Model
  - Background
  - Benefits / Criticisms
  - Updates: Hot Cheese Model
- Real Disasters
  - Deepwater Horizon
  - Challenger
- Conclusion

Swiss Cheese Model

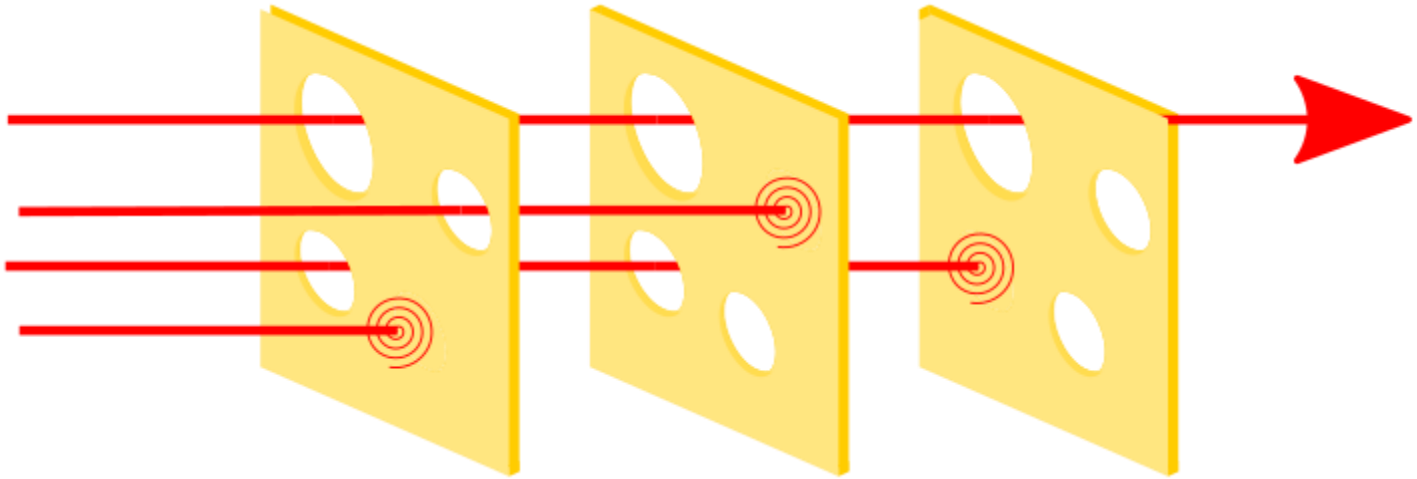
Swiss Cheese Model – History

- The Swiss Cheese Model is a way to understand how failures happen in complex systems

- Introduced by James Reason ("Human Error," 1990)

- Layered prevention (slices of cheese)
- Each layer has vulnerabilities (holes in cheese)
- Accidents happen when:
  - Barrier does not exist (holes align)
  - Barriers fail (slice is removed, holes align)

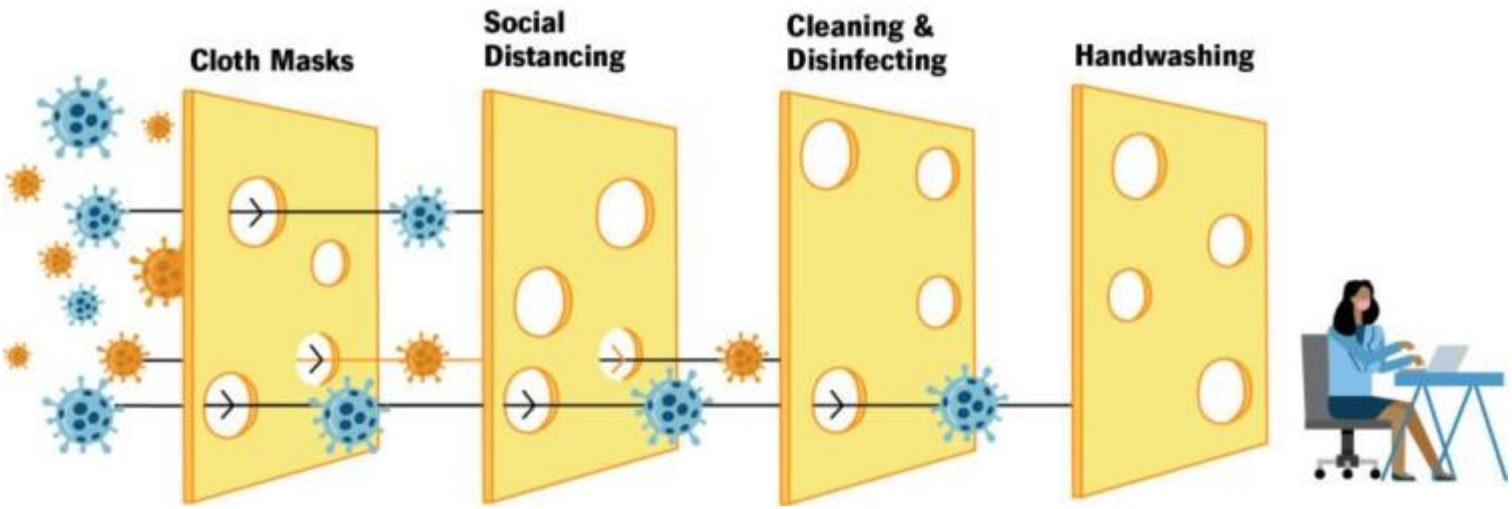
Swiss Cheese Model – Graphic



Source: By User:BenAveling - File:Swiss cheese model.svg, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=133912327

Swiss Cheese Model – Example

Preventing the spread of COVID-19



Source: https://www.linkedin.com/pulse/layered-defenses-addressing-holes-how-swiss-cheese-model-sam-lindgren/

Swiss Cheese Model – Applications

- Used in multiple high-risk, high-complexity industries, such as:
  - Aviation
  - Nuclear Power
  - Healthcare

Swiss Cheese Model – Benefits

- Simplicity
  - Framework to understand risk reduction

- Graphical
  - Highlights there are many approaches to prevent harms (e.g., adding a slice, reducing hole size)

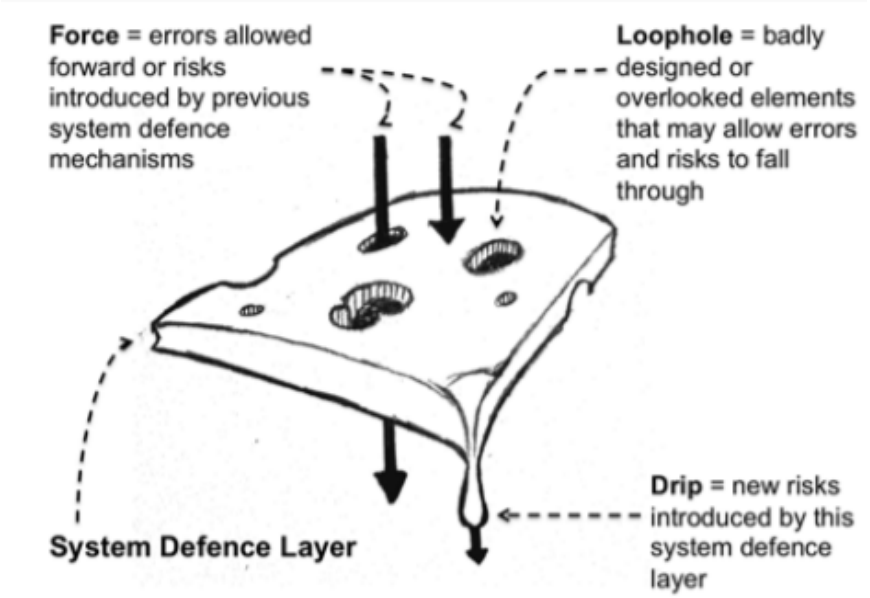
Swiss Cheese Model – Criticisms

- Simplicity
  - Assumes independence of the barriers, static system
  - Focuses on implementing barriers to prevent issues, rather than addressing the underlying causes of the issues themselves

- Generic
  - Open to interpretation – creates confusion, disagreement
  - Limited practicality

Swiss Cheese Model – Updates

- Hot Cheese Model
  - Introduced by Li & Thimbleby (2014)
  - Dynamic (melting cheese)
    - Barriers change over time (new holes)
    - Poorly designed mitigations (holes change shape)
    - Each barrier can introduce new risks (drip)



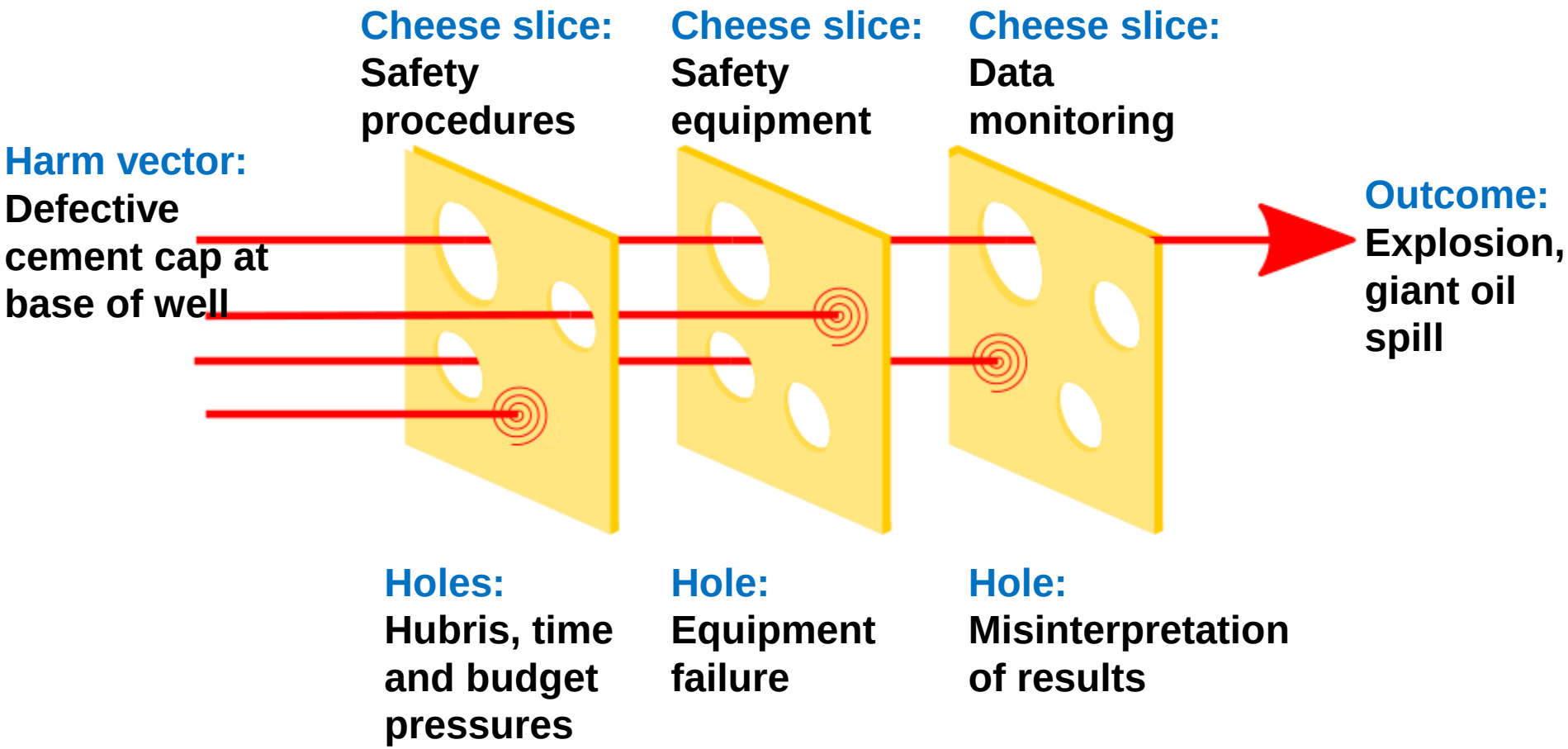
Source: Li Y & Thimbleby H. Hot cheese: a processed Swiss cheese model. *J R Coll Physicians Edinb* 2014; 44:116–21 http://dx.doi.org/10.4997/JRCPE.2014.205

Swiss Cheese Model – Summary

- Model used in risk management based on the principle of layered security
- Helps reduce risk of accidents occurring and / or reduce impact of accidents when they occur
- Helps us understand how accidents happen

Real Disasters

Deepwater Horizon Disaster – Swiss Cheese Model

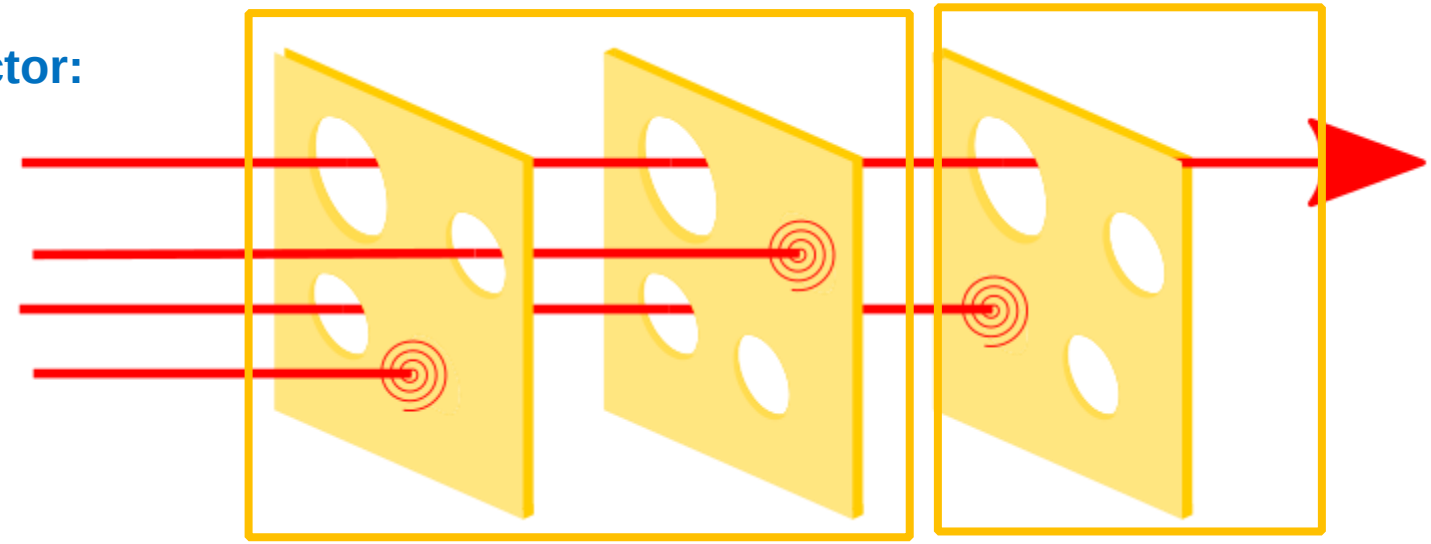


Challenger Disaster – Swiss Cheese Model

Cheese pack:
Design & Testing

Cheese pack:
Oversight / Culture

Harm vector:
O-rings failed



Outcome:
Shuttle explosion

Holes:
- Material impacted by cold
- Redundant systems had same material properties

Hole:
Prioritized meeting the launch schedule over safety

Conclusion

- Major accidents and catastrophic systems failures tend to be caused by multiple, smaller failures and hazardous situations leading up to the actual harm

- Identifying and mitigating hazards or potentials for error can prevent catastrophes and save lives

- Understanding risk mitigation by using models for analysis, like the Swiss Cheese Model, can help reduce risk of accidents occurring and / or reduce impact of accidents when they occur